Supplementary Materials for Disturbances Robust Mamba Model for Industrial Anomaly Diagnosis

IV. Experimental Studies

B. Industrial IPAD Example

4) Temporal Dependency Analysis: To investigate the influence of temporal information on dynamic feature extraction, we conducted comparative experiments by replacing the temporal cues in the preprocessing stage with constant values, thereby weakening temporal information dependency. The experimental results are shown in Fig. S1 and Table SI.

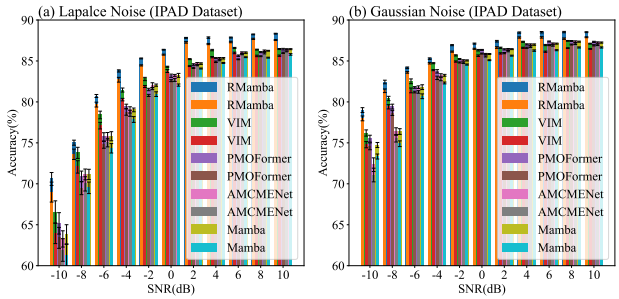


Fig. S1. The accuracy of RMamba under noise with different SNR with and without temporal information.

TABLE SI. Influence of temporal information on the average classification accuracy of different models under Laplace and Gaussian noise on the IPAD dataset, where $Accuract_{N1}$ and $Accuracy_{N2}$ denote the results without temporal information, while $Accuracy_1$ and $Accuracy_2$ denote the results with temporal information.

Model	Accuracy$_{N1}$ (%) → Accuracy$_1$ (%)	Accuracy$_{N2}$ (%) → Accuracy$_1$ (%)
Mamba	79.224±0.337→80.346±0.305	82.945±0.193→83.840±0.140
PMOFormer	79.763±0.346→80.366±0.306	83.168±0.190→84.328±0.203
AMCMENet	79.609±0.318→80.294±0.319	83.373±0.140→83.622±0.135
VIM	80.557±0.297→81.661±0.305	83.794±0.151→84.666±0.136
RMamba	**82.918±0.269→83.810±0.167**	**85.300±0.116→86.033±0.114**

It can be observed that Mamba-based models exhibit a more obvious performance degradation after removing temporal information, while CNN-based and Transformer-based models show relatively smaller decreases. This is because Mamba-based models rely more heavily on temporal cues and state-space modeling to capture dynamic evolution patterns. When temporal information is weakened, their ability to model sequential state transitions is directly affected. In contrast, CNN-based models mainly depend on local spatial texture features, and Transformer-based models primarily capture global spatial dependencies. Therefore, their performance is less sensitive to the removal of temporal information.

Although RMamba also shows a noticeable decrease when temporal information is removed, it still achieves the highest accuracy among all compared models. This advantage benefits from the proposed temporal-aware attention Mamba and SSM-based dynamic representation mechanism, which can still extract latent dynamic characteristics from feature-level responses. Meanwhile, the GF enhances global discriminative representations, and the DGM suppresses disturbance-dominated redundant responses. Therefore, RMamba maintains superior robustness and diagnosis accuracy even when explicit temporal information is weakened.

5) Ablation Experiment: To analyze the contribution of each component in RMamba, ablation experiments are conducted by modifying different modules in the proposed framework, and the experimental results are shown in Fig. S2 and Table SII. According to the robustness evaluation results presented in the previous subsection, Gaussian noise and Laplace noise provide representative and controllable perturbation conditions for robustness analysis. Therefore, the ablation experiments are conducted under Gaussian and Laplace noise settings to further evaluate the contribution of each module under different disturbance intensities.

Firstly, the traditional Mamba framework is adopted instead of the temporal-aware attention Mamba framework, which is referred to as RAMamba. The experimental results show that the temporal-aware attention Mamba, as a parallel framework,

only requires a slight increase in computational cost while significantly enhancing the model's dynamic feature extraction capability.

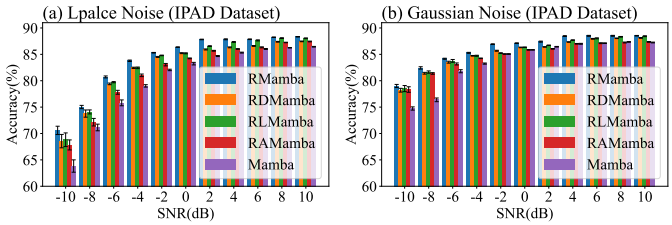


Fig. S2. Diagnosis accuracy of the ablation experiment model under Laplace noise and Gaussian noise in the SNR range from -10 dB to 10 dB (IPAD dataset).

TABLE SII. Comparison results of RMamba ablation experiments. $Accuracy_1$ denotes the average accuracy under Laplace noise on the IPAD dataset, while $Accuracy_2$ denotes the average accuracy under Gaussian noise on the IPAD dataset.

Model	Parames(k)	MFLOPs	Accuracy$_1$(%)	Accuracy$_2$(%)
Mamba	**213.031**	**23.181**	80.346±0.305	83.840±0.140
RAMamba	**213.031**	**23.181**	81.710±0.268	84.802±0.129
RLMamba	214.470	**23.181**	82.978±0.225	85.406±0.147
RDMamba	214.470	23.184	82.511±0.255	85.247±0.112
RMamba	**214.470**	**23.181**	**83.810±0.167**	**86.033±0.114**

Secondly, the lightweight global feature extraction module is removed, referred to as RLMamba. Experimental results show that this module can effectively enhance the global representation capability of industrial dynamic features with only a small number of learnable parameters, thereby improving diagnosis accuracy and robustness without introducing excessive computational overhead.

Finally, the dynamic gating module is removed, referred to as RDMamba. Experimental results demonstrate that the dynamic gating module can effectively suppress disturbance-dominated feature responses under strong disturbance conditions, thereby improving the robustness of the proposed framework while maintaining computational efficiency.

C. Industrial FMF Example

4) Temporal Dependency Analysis: To further verify the influence of temporal information on the FMF dataset, we also conducted temporal information analysis experiments under FMF conditions by replacing the temporal cues with constant values. The experimental results are shown in Fig. S3, Fig. S4, Table SIII, and Table SIV. The experimental results show that a similar phenomenon can be observed on the FMF dataset. Specifically, after temporal information is weakened, Mamba-based models show a more obvious performance degradation than CNN-based and Transformer-based models. This is because Mamba-based models rely more strongly on temporal cues and state-space modeling to capture the dynamic evolution characteristics of industrial processes. When temporal information is removed, their sequential dynamic modeling capability is directly affected.

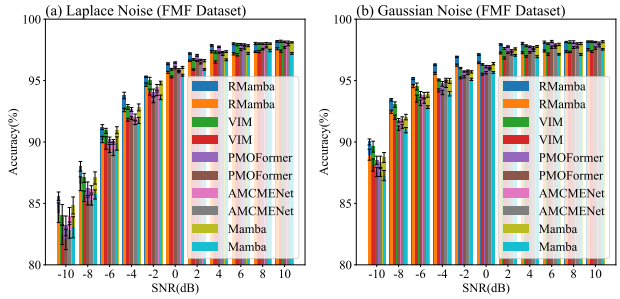


Fig. S3. Influence of temporal information on the robustness of different models under Laplace and Gaussian noise on the FMF classification task.

TABLE SIII. Influence of temporal information on the average classification accuracy of different models under Laplace and Gaussian noise on the FMF dataset, where $Accuract_{N1}$ and $Accuracy_{N2}$ denote the results without temporal information, while $Accuracy_1$ and $Accuracy_2$ denote the results with temporal information.

Model	Accuracy$_{N1}$ (%) → Accuracy$_1$ (%)	Accuracy$_{N2}$ (%) → Accuracy$_2$ (%)
Mamba	93.087±0.242→94.029±0.212	94.686±0.131→95.559±0.152
PMOFormer	93.232±0.215→93.670±0.191	95.115±0.139→95.397±0.147
AMCMENet	93.164±0.202→93.667±0.220	94.949±0.138→95.416±0.131
VIM	93.034±0.237→94.036±0.216	94.933±0.146→95.763±0.140
RMamba	**93.642±0.214→94.506±0.154**	**95.548±0.122→96.303±0.089**

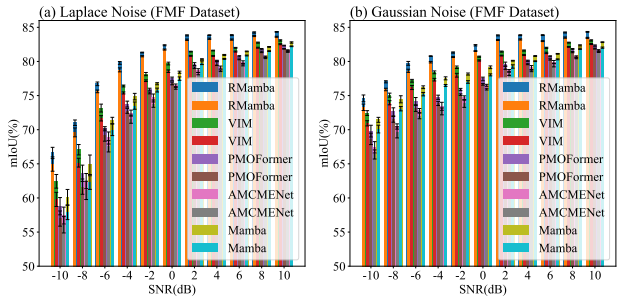


Fig. S4. Influence of temporal information on the robustness of different models under Laplace and Gaussian noise on the FMF segmentation task.

TABLE SIV. Influence of temporal information on the average segmentation mIoU of different models under Laplace and Gaussian noise on the FMF dataset, where $mIoU_{N1}$ and $mIoU_{N2}$ denote the results without temporal information, while $mIoU_1$ and $mIoU_2$ denote the results with temporal information.

Model	mIoU$_{N1}$ (%) → mIoU$_1$ (%)	mIoU$_{N2}$ (%) → mIoU$_2$ (%)
Mamba	74.702±0.406→75.816±0.380	77.579±0.196→78.559±0.179
PMOFormer	73.128±0.472→73.845±0.479	75.329±0.351→75.934±0.314
AMCMENet	74.099±0.401→74.930±0.380	76.475±0.261→77.203±0.309
VIM	75.945±0.261→77.073±0.297	78.569±0.171→79.437±0.158
RMamba	**78.957±0.254→79.810±0.199**	**80.560±0.136→81.446±0.138**

Nevertheless, RMamba still maintains clear advantages in both classification accuracy and segmentation mIoU on the FMF dataset. This is mainly attributed to the proposed temporal-aware attention Mamba, which enhances dynamic feature extraction under limited temporal information. Meanwhile, the GF improves global discriminative representation capability, and the DGM suppresses disturbance-dominated redundant responses. Therefore, RMamba can still achieve superior robustness and diagnosis performance even when explicit temporal information is weakened.

5) Ablation Experiment: To further evaluate the contribution of each component on the FMF dataset, ablation experiments are conducted for both classification and segmentation tasks. According to the robustness evaluation results presented in the previous subsection, Gaussian noise and Laplace noise provide representative and controllable perturbation conditions for robustness analysis. Therefore, the ablation experiments are conducted under Gaussian and Laplace noise settings, with the classification results shown in Table SV and Fig. S5, and the segmentation results shown in Table SVI and Fig. S6. The experimental results demonstrate that the proposed framework maintains stable effectiveness across different industrial tasks and disturbance conditions.

TABLE SV. Comparison results of RMamba ablation experiments. $Accuracy_1$ denotes the average accuracy under Laplace noise on the FMF dataset, while $Accuracy_2$ denotes the average accuracy under Gaussian noise on the FMF dataset.

Model	Parames(k)	MFLOPs	Accuracy$_1$(%)	Accuracy$_2$(%)
Mamba	**52.681**	**0.321**	94.029±0.212	95.559±0.152
RAMamba	**52.681**	**0.321**	94.040±0.213	95.728±0.143
RLMamba	55.216	**0.321**	94.461±0.181	96.275±0.104
RDMamba	55.216	0.322	94.165±0.184	96.051±0.133
RMamba	**55.216**	**0.321**	**94.506±0.154**	**96.303±0.089**

TABLE SVI. Comparison results of RMamba ablation experiments. $mIoU_1$ denotes the average mIoU under Laplace noise on the FMF dataset, while $mIoU_2$ denotes the average mIoU under Gaussian noise on the FMF dataset.

Model	Parames(k)	MFLOPs	mIoU$_1$(%)	mIoU$_2$(%)
Mamba	**98.136**	**0.675**	75.816±0.380	78.559±0.179
RAMamba	**98.136**	**0.675**	77.718±0.307	79.728±0.154
RLMamba	103.866	**0.675**	78.764±0.257	80.495±0.137
RDMamba	103.866	0.676	76.672±0.301	79.578±0.145
RMamba	**103.866**	**0.675**	**79.809±0.199**	**81.446±0.138**

Specifically, in the classification task (Table SV), RMamba achieves the highest average accuracy under both Laplace noise and Gaussian noise, reaching 94.506 % and 96.303 %, respectively. This demonstrates that the proposed modules effectively improve robustness under both impulsive and continuous disturbance conditions.

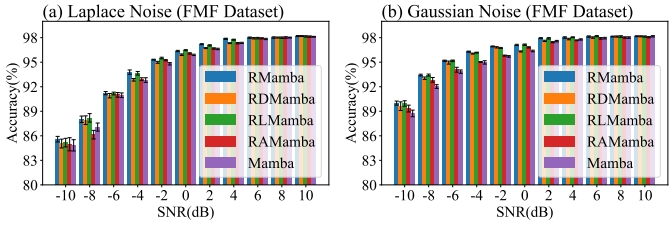


Fig. S5. Diagnosis accuracy of the ablation experiment model under Laplace noise and Gaussian noise in the SNR range from -10 dB to 10 dB (FMF dataset).

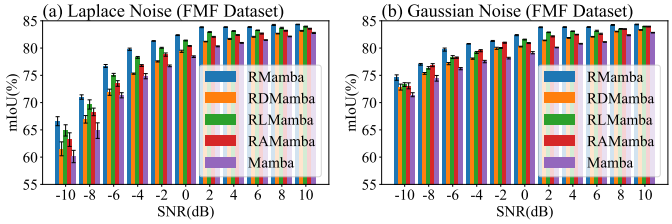


Fig. S6. Segmentation mIoU of the ablation experiment model under Laplace noise and Gaussian noise in the SNR range from -10 dB to 10 dB (FMF dataset).

In the segmentation task (Table SVI), RMamba also achieves the best performance, with mIoU values of 79.810 % under Laplace noise and 81.446 % under Gaussian noise, indicating that the proposed framework can effectively preserve semantic consistency and segmentation quality under noisy industrial environments.

Furthermore, Fig. S5 and Fig. S6 further visualize the robustness advantages of RMamba across the full SNR range. As the SNR decreases, the performance gap between RMamba and other variants becomes increasingly significant, especially under low-SNR conditions such as -10 dB and -8 dB. These results further demonstrate the effectiveness of the proposed modules in improving both classification and segmentation robustness under complex industrial disturbance environments.